cragshift: Quality-aware camera–laser monitoring with an auditable simulated rock-slope benchmark

Reproducible research instrument and experiment report

13 July 2026 Simulation study; no field-safety claim

Abstract

We build a reproducible camera–laser inspection instrument and test learned quality-dependent fusion on original rendered surveys, independently raycast laser observations and withheld simulator truth. Two nonlinear residual-scale models weight commensurate surface-normal evidence with an uncertainty bound that does not assume independent sensor errors. An initially proposed temporal confirmation rule fails: it misses early movement. After preserving that result, we fix the simpler method and replicate it on sixteen newly seeded sites, without changing weights or calibration thresholds. Recall is 82.33% versus 75.93% for simple fusion, at observed-negative false-positive rates 5.60% and 5.42%, respectively, with identical 95.13% observation coverage. The paired whole-site bootstrap recall difference is 6.39 percentage points, with interval [4.28,8.86]. A descriptive equal-FPR comparison also favors learned fusion, but it does not establish deployment calibration. Source, native scenes, observations, weights, raw results, failed alternatives, independent reviews, an offline inspection app and original footage are retained. The result is a tested simulated monitoring instrument, not a physically calibrated rockfall predictor or field validation.

1 Research question and scope

Repeated surveys must distinguish measured movement from misregistration, visibility changes, weak image tracks and sparse scans. The question is whether a removable learned quality model improves detection over simply combining sensors, while preserving coverage and uncertainty evidence. We study observed normal-component motion and candidate inspection regions. Surface separation, tracked material displacement, completed rockfall and future instability remain distinct; no collapse horizon is predicted.

The engineering contribution is an end-to-end, provenance-preserving instrument. The tested mechanism is quality-conditioned residual scaling with dependence-conservative weighting. It is identifiable in source, exportable as numeric weights and removable in comparisons. We do not claim that uncertainty weighting or neural fusion is unprecedented. M3C2 and M3C2-EP already address geometric change and uncertainty [1, 2]; learned reliability has precedents in remote-sensing fusion, including HAEFNet [12]. Our evidence supports the specified simulated task, not a universal novelty or real-slope reliability claim.

A public-data audit found useful but incomplete field components. The Trier release supplies an actual before/after laser pair; HovoPhoto40 supplies calibrated multi-epoch images, marked coordinates and total-station references, but its public dense clouds are photogrammetric [4, 5, 6]. They are different sites and are never paired as multimodal data. After that access finding, the project scope explicitly changed to simulation. All synthetic observations and results are labeled accordingly; the separate field analyses remain available as component checks.

2 Original survey benchmark

2.1 Scenes, acquisition and truth separation

Blender 5.1.2 generates original convex outcrop fragments from irregular clipped Voronoi cells, open joints, a nonrectangular crest, variable thickness, procedural mineral texture and talus. No external visual assets or videos are copied. The first experiment uses 24 training, 8 calibration and 8 held-out procedural sites. A later independent replication uses 16 fresh seeds, two per nuisance condition. The development smoke site is excluded from every split. Entire sites and all their epochs remain together; overlapping anchor patches are never randomly split.

Each site has six instantaneous surveys, calibrated 640×480 stereo cameras with a 0.6 m baseline, and a separate 128×96 first-hit laser ray grid. Camera coordinates use OpenCV's right/down/forward convention, while the world uses metres with Z up. Camera matrices, scanner origin, units, acquisition indices and time definition are explicit. Prescribed creep translates selected fragments by a site-dependent 22–55 mm per synthetic survey interval, with smaller lateral/vertical components. These artificial seconds are survey indices, not a geological creep-rate assertion or a failure law.

Laser rays intersect evaluated scene meshes. Angular jitter and Gaussian range noise are applied independently of rendered image formation; one shared pose perturbation per survey can affect both nominal sensor reconstructions. Range standard deviation is the declared heuristic $0.008(1 + 0.25r/20)$ m, angular jitter is 0.0003 rad, and dropout depends on range and incidence. These are implemented sensitivity models, not parameters fitted to a commercial scanner. They follow the distinction between geometric ray casting, sensor noise and platform errors emphasized in sensor-simulation work [7, 8].

Conditions are clean, changed shadows, transient vegetation, foreground occlusion, sparse scans, shared misregistration, missing imagery and missing laser observations. Missing-modality cases preserve epoch zero so every method has the same initial query anchors. The simulation validates nearest-hit occlusion with an analytic near/far slab fixture. Camera projection agrees with Blender's independent camera projection within 0.000046 pixels on the development scene; implemented standardized range noise has approximately zero mean and unit variance. These are implementation checks, not physical sensor calibration.

Observation files contain rendered RGB, noisy returns, emission-ray indices and nominal calibration. Separate truth files contain object identities, true poses, clean ranges, transformed material anchors and first-hit visibility. The extractor has no truth-file interface. Tests demonstrate that poisoned/nonexistent truth paths do not affect extraction, and changing the current laser cloud leaves image estimates bit-identical. Training alone may read training-site truth. The evaluator maps selected noisy reference returns to truth through exact emitted-ray lineage, avoiding guessed nearest-surface labels.

2.2 Comparable evidence and registration

The extractor selects 220 actual epoch-zero laser returns per site, estimates sensor-facing local normals and defines shared query neighborhoods. Each later laser observation is aligned using declared left/bottom static structural reference bounds. The image branch separately estimates its stable-control registration. Registration uses no moving-region labels; fit residuals, covariance approximations and failures are retained. Local alignment uncertainty is projected at each anchor, including rotational lever arm. Correlated baseline errors are not reduced by repeatedly observing the same reference.

The geometry measurement is the difference of projected local cylindrical means, with 0.24 m radius, 0.65 m axial half-length and at least six returns. Spread and foreground/occlusion checks can reject a comparison. Image features are detected near projected reference anchors, matched in stereo and tracked across time with forward/backward, reprojection, ray-angle and photometric checks. Current and reference triangulated coordinates are differenced and projected onto the initial normal. Current laser values never enter image displacement. Initial laser anchors and nominal calibration are shared context, so image and laser estimates are not independent confirmations by construction.

Both measurements approximate the same normal component where a common rigid material patch remains observable. Mixed neighborhoods, discontinuities and occlusion can violate this approximation; their errors and rejections remain in evaluation. A rejected or unseen surface is unknown, not stable. The method does not convert normal separation into an arbitrary three-dimensional displacement vector.

3 Learned quality fusion and the failed temporal rule

Two small ReLU networks predict log absolute residual scale from measured diagnostics. Geometry inputs include return counts, local spread, absolute incidence cosine and registration residual; image inputs include accepted-track count, forward/backward and reprojection error, gradient, triangulation angle, normal-track dispersion and registration residual. No object ID, motion stage, nuisance-condition label, true pose or future epoch is a feature. Network sizes are 481 and 529 parameters, trained on 15,294 geometry and 21,968 image observations from training sites only. Standardization is training-only. The target is $\log \max(|\hat{d} - d^*|, 0.002)$; conversion by 1.887 is a Gaussian-reference scale convention, not proof that residuals are Gaussian. Scales are clipped and bounded below by shared-reference uncertainty. Optimizer warnings, including the image model's fixed iteration budget, are preserved.

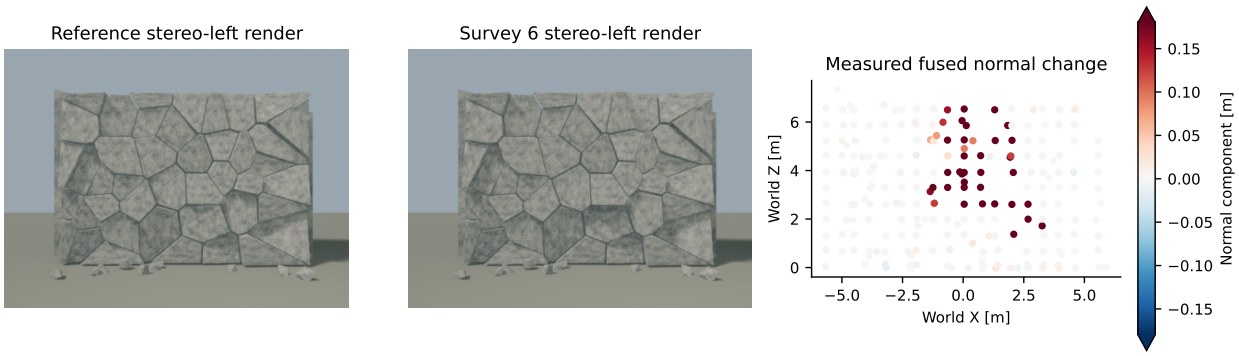


Figure 1: Original development-site observations and observation-only fused evidence. Photographs are Blender renders; laser data come from separate ray casts. The color plot shows normal-component estimates on measured initial anchors, not a hazard map or simulator truth.

For available channel estimates d_k and scales s_k, weights are positive and normalized:

$$w_k = \frac{s_k^{-2}}{\sum_j s_j^{-2}}, \qquad \hat{d} = \sum_k w_k d_k, \qquad s_{\mathrm{dep}} = \sum_k w_k s_k.$$

For fixed weights, s_{dep} bounds the standard deviation of this convex combination under arbitrary cross-channel dependence when marginal standard deviations are valid. Because our weights depend on observed diagnostics, applying that covariance Cauchy–Schwarz argument requires conditioning on the estimated weights and measured diagnostics. Conditional marginal scales must also be valid. Those assumptions are not guaranteed by residual-scale learning; the resulting intervals require empirical checking. We therefore report conditional coverage and interval width rather than asserting a distribution-free guarantee. A single available modality is retained; no available estimate yields abstention. The score is $|\hat{d}|/s_{\mathrm{dep}}$. Displayed intervals have radius $1.96 s_{\mathrm{dep}}$.

The initial temporal rule required same-sign prior corroboration and used the minimum of current and recent evidence. It did not shrink interval widths, but it necessarily missed the first movement observation. An earlier four-state correlated-likelihood proxy experiment also failed its simpler likelihood ablation; it and its shortcut challenges are retained separately. The rendered-sensor experiment supplies a more substantive test, yet its confirmation rule fails too. We preserve those failures and adopt the simpler current-observation decision score. Temporal region association remains an inspection function based on overlap of fixed observed anchor sets; it is not extra independent measurement precision.

4 Evaluation and selection boundary

The first eight-site held-out test diagnosed the confirmation failure: learned fusion with confirmation recalled 67.96%, simple fusion 77.57%, and the no-confirmation learned variant 82.44%. These numbers cannot independently validate a choice made after examining them. We therefore fixed the simpler variant in a new replication protocol, reserved fresh seeds 400–415, and retained the exact same trained weights and original calibration thresholds. No threshold or weight was changed using replication outcomes.

All nine methods receive the same observations and initial queries: image only, geometry only, simple equal-weight independent-error fusion, nominal-scale dependence-conservative fusion, learned fusion and its temporal/quality/dependence variants, plus actual upstream py4dgeo 1.2.1 M3C2 [3]. M3C2 uses identical query normals, registration and geometric rejection gates; raw distances agree with the custom cylinder estimator to 8.57×10^{-16} m on the smoke case. This is an implementation-equivalence check, not a new M3C2 algorithm. Ordinary M3C2 uncertainty includes the same projected alignment approximation.

We do not label that approximation M3C2-EP. The installed EP API's scanwise range/yaw/scan-angle noise model cannot faithfully represent the simulator's varying per-return range noise and isotropic direction jitter without approximation. Its affine covariance and three-degree-of-freedom detection threshold also differ. The exact mismatch is documented; no EP-equivalence claim is made. The published PointNet/PointNet++ completed-rockfall baseline [11] requires labeled deformation-cluster meshes and its original CUDA/TensorFlow stack. It was inspected but not executed or replaced by another model under its name. It is not a comparator for future collapse.

Method	Recall [%]	FPR_{obs} [%]	Coverage [%]	RMSE [mm]	Interval coverage [%]
Geometry	52.39	5.60	57.66	13.24	97.45
Image	56.97	5.25	83.87	38.58	98.78
Simple fusion	75.93	5.42	95.13	25.17	98.54
Analytic + confirmation	63.60	5.35	95.13	20.40	98.03
Learned + confirmation	67.13	4.92	95.13	20.17	98.59
Constant scales + confirmation	65.74	5.79	95.13	20.48	95.17
Learned quality fusion	82.33	5.60	95.13	20.17	98.59
Independent errors + confirmation	66.92	5.13	95.13	20.17	98.22
Upstream M3C2	51.52	5.31	57.66	13.24	96.49

Table 1: Fresh sixteen-site test, fixed calibration thresholds. Recall retains abstentions. Error/interval columns use each method's observed visible subset; fusion methods have equal coverage, while individual sensors do not. Actual test FPRs differ.

Method	Matched	False	Missed	Precision [%]	Recall [%]
Simple fusion	65	67	15	49.24	81.25
Learned quality fusion	72	60	8	54.55	90.00
Learned + confirmation	58	43	22	57.43	72.50
Geometry	51	66	29	43.59	63.75
Image	47	52	33	47.47	58.75
Upstream M3C2	50	60	30	45.45	62.50

Table 2: Secondary normal-motion region counts across 80 simulated intervals, using the fixed region protocol. These are not labeled completed rockfalls.

Each score threshold targets 5% false positives among observed unchanged calibration anchors. Epoch-zero self-differences are excluded. Known normal motion of at least 10 mm is positive. Recall includes all positive anchor-epochs, including abstentions; false-positive rates are reported against both all negatives and observed negatives. Error and interval coverage use observed estimates of currently visible material. Thus lower single-channel errors on smaller observed subsets are not directly comparable with fusion's broader coverage.

Secondary region metrics connect at least three detected anchors within 0.77 m, match truth/prediction regions one-to-one at anchor-set IoU ≥ 0.25, and retain unmatched truth regions as misses. False regions per projected area use the declared $11.2 \times 7.6 = 85.12$ m^2 scanner target-plane window. This is not rough three-dimensional surface area, completed-rockfall event exposure or a field false-alarm rate. Whole-site paired bootstrap resampling, 2,000 draws, retains within-site dependence. Sixteen procedural sites remain a small sample from one generator.

5 Fresh replication results

The selected learned quality fusion recalls 82.33% versus 75.93% for simple fusion, at observed-negative FPR 5.60% versus 5.42% and identical 95.13% coverage (Table 1). Across 17,600 known anchor-epochs there are 2,863 positives. Paired whole-site bootstrap gives recall difference 6.39 percentage points [4.28,8.86]. The observed-negative FPR difference is +0.177 percentage points, with interval $[-0.931, +1.601]$. This is not evidence of identical fixed-test FPR or a formal noninferiority result.

At an exactly shared descriptive all-negative FPR of 4.994%, learned fusion recalls 81.73% and simple fusion 75.24% (Fig. 2). These thresholds describe held-out score ranking; they are not substituted for the frozen operating thresholds. The specified curve points also favor learned fusion at 1% and 2.5%. Same-coverage visible-component RMSE falls from 25.17 to 20.17 mm. Mean nominal 95% interval width decreases from 143.64 to 92.90 mm, with empirical coverage 98.54% and 98.59%, respectively. These intervals are conservative in this simulation, not exactly calibrated 95% field intervals.

Region recall is 90.00% versus 81.25%, precision 54.55% versus 49.24%, and false-region density 0.881 versus 0.984 per 100 projected m^2 per interval (Table 2). Region precision remains modest. The confirmation variant has fewer false regions but more misses, so it is not universally dominated. Robustness includes missing modalities and artifact cases (Fig. 3); only two replication sites represent each condition, so nuisance effects and site geometry remain partly confounded. We do not treat 17,600 correlated anchor-epochs as 17,600 independent trials.

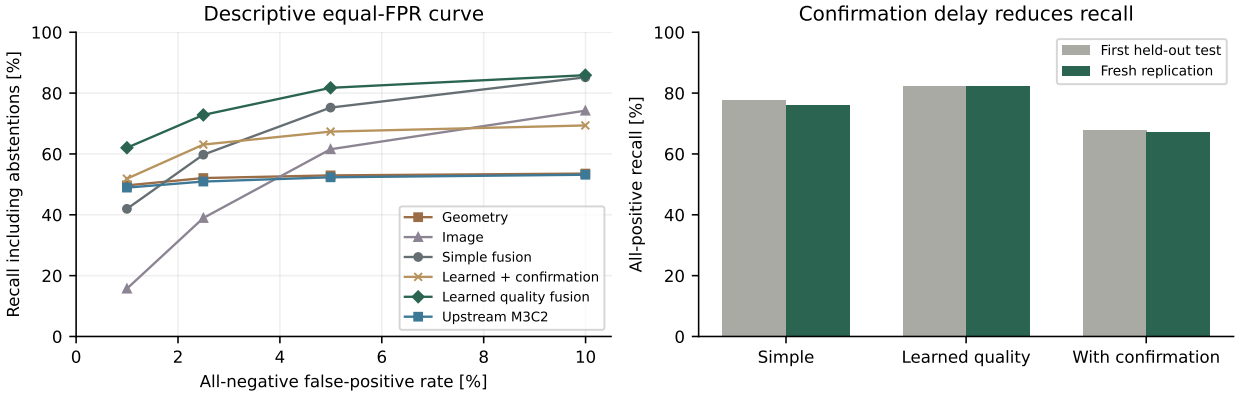


Figure 2: Left: descriptive equal-FPR comparisons retain every positive in recall. Right: both held-out experiments preserve the failure of confirmation delay. The revised method's independent evidence is the fresh replication, not the first test used in its selection.

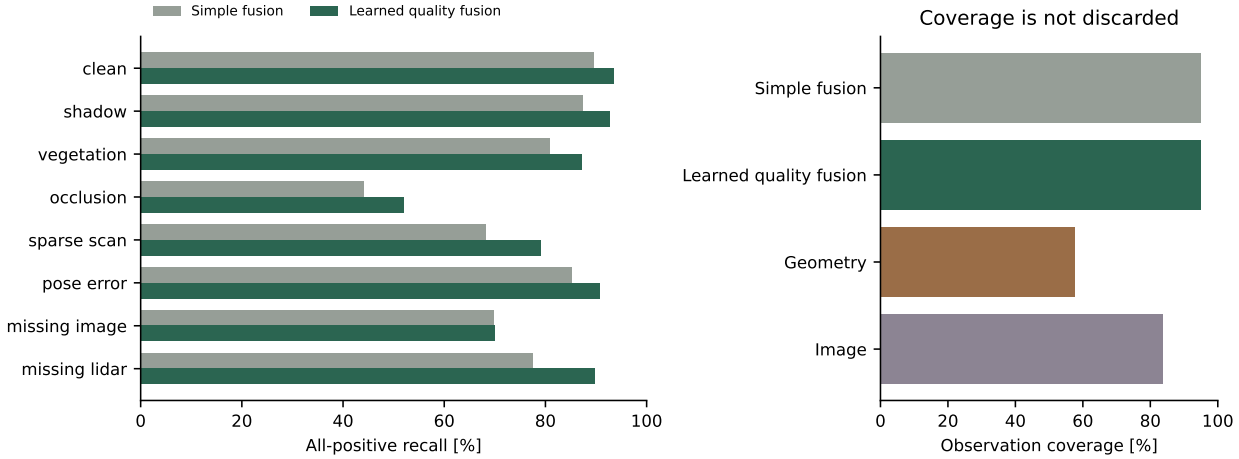


Figure 3: Controlled-condition sensitivity and observation coverage. Same fusion coverage prevents attributing its gain simply to rejecting more difficult scenes. Conditions have two sites each; the plot does not establish population-level robustness.

6 Physics checks and original demonstration

Prescribed survey creep is separate from gravity-driven post-release dynamics. Real experimental literature motivates explicit shape, rotation, substrate and fragmentation limits [9, 10]; it does not calibrate our materials. Original footage uses the same original convex meshes, with prescribed initial release positions, in a controlled flat-ground drop.

Native Blender Bullet passes gravity and sliding checks but fails restitution/step-sensitivity checks; that failure is retained. Free MuJoCo 3.17.0 passes fifteen selected sphere/slide fixtures and seven timestep comparisons using an analytically derived unilateral compliant-contact damping relation. No measured rock coefficient is asserted. Transfer to the original rough cliff shows nonconverged late runout, retained as a failed scope extension. A separate farther-forward controlled drop isolates ground contact. Halving timestep from 5×10^{-5} to 2.5×10^{-5} s changes first-impact time by at most 25 microseconds, first center-of-mass rebound apex by 0.545%, and three-second runout by 0.370% and 1.540% for the two bodies. Absolute compliant penetration remains about 14.15 mm and fails the earlier 12 mm sphere comparison. This is explicitly a passed first-impact subset, not a calibrated fragmentation/runout or failure-onset model.

Blender renders imported position/quaternion trajectories with its Bullet world disabled. Native pose roundtrip checks, source scenes, XML, full-rate states, timing and video decode evidence are retained. All footage is original; external experiment videos are cited as source pointers, not copied footage or claimed measured trajectories. The observation benchmark depends on prescribed transforms, so the separate contact-model failure does not silently alter its motion truth.

7 Inspection, reproduction and limits

The independent CLI validates survey metadata and checksums, aligns references, estimates change and exports results without the interface. The offline app shows six synchronized stereo surveys, before/after laser support, normal-component evidence, candidate-region timelines, per-channel values, conditional uncertainty and complete JSON/CSV exports. Invalid imports preserve prior work. Missing imagery/geometry remains visible as unknown. Actual desktop/mobile browser checks and screen recordings are retained.

Another researcher can regenerate the 56 procedural sites, extract observations without truth access, load numeric weights, reproduce frozen calibration and both held-out evaluations, inspect raw results and compile this LaTeX paper with result-generated figures. Source snapshots, hashes, pinned dependencies, test results and independent numerical reviews are preserved. The field-component audit, failed four-state proxy, failed temporal rule, failed native bounce and rough-cliff dynamics are not discarded.

The main limits are one procedural generator, heuristic sensors, ideal nominal camera calibration apart from declared pose tests, prescribed creep, small site counts, mixed-patch observability and incomplete physical contact transfer. A faithful learned completed-rockfall baseline and matched-noise M3C2-EP run remain unavailable. Scientific priority or broad novelty is not established. The supported conclusion is improved simulated normal-motion ranking and a useful inspectable research instrument; real paired surveys and independently measured physical sensor/trajectory errors are still required for field or operational claims.

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